

Detailed Documentation: How to Create an EC2 Instance in AWS

Amazon EC2 (Elastic Compute Cloud) is a core AWS service that allows you to launch and manage virtual servers (instances) in the cloud. Below is a comprehensive guide covering all steps, their purpose, and best practices.

1. Prerequisites

Before launching an EC2 instance, ensure you have:

- An **AWS Account** (with valid payment method)
- Basic understanding of:
 - Regions & Availability Zones
 - Networking (VPC, Subnets)
 - SSH / Key Pairs
- IAM permissions:
 - `ec2:RunInstances`
 - `ec2:Describe*`
 - `ec2:CreateKeyPair`
 - `ec2:CreateSecurityGroup`

2. Step-by-Step Process to Launch an EC2 Instance

Step 1: Log in to AWS Management Console

1. Open: <https://console.aws.amazon.com>
2. Sign in using:
 - Root Account (not recommended)
 - IAM User (recommended)
 - IAM Role (when using AWS SSO)

Step 2: Choose a Region

Select the region from the top-right corner.

Why important?

- EC2 resources are *region-specific*.
- Pricing varies by region.
- Minimizes latency for users near the region.

Example: **US East (N. Virginia)**, **Asia Pacific (Mumbai)**.

Step 3: Open the EC2 Dashboard

From the AWS Console:

1. Click **Services**
2. Search for **EC2**
3. Open the EC2 Dashboard

You will see:

- Running Instances
- Elastic IPs
- Security Groups
- Key Pairs
- Volumes

Step 4: Click "Launch Instance"

This starts the configuration wizard.

3. Configure the Instance

Step 5: Name and Tags

Set a name for the instance:

Name: MyFirstEC2

Environment: Development

Project: WebApp

Tags help in:

- Billing
- Organization
- Automation (e.g., AWS Lambda scripts)

Step 6: Choose an AMI (Amazon Machine Image)

Examples of common AMIs:

AMI	Description
Amazon Linux 2023	Good for general workloads
Ubuntu 22.04 LTS	Popular for development
RedHat Enterprise Linux	Enterprise-grade
Windows Server	GUI-based workloads
Custom AMI	Your own pre-configured templates

Choose **Amazon Linux 2023** for practice (Free-tier eligible).

Step 7: Choose an Instance Type

Instance type determines CPU, RAM, and network performance.

Common free-tier type:

- **t2.micro** (1 vCPU, 1 GB RAM)
- **t3.micro** (better performance if credits exist)

Examples:

Use Case	Instance Type
Web App	t2.micro / t3.micro
High CPU	c5.large
High Memory	r6g.large
Machine Learning	g4dn.xlarge

Step 8: Create / Select a Key Pair

Key pair is used for SSH/RDP access.

- Name: **my-key**
- Type: RSA (for Linux)
- File Format: **.pem** (Mac/Linux/Windows) or **.ppk** (Windows Putty)

Important

Save it securely; AWS will never show it again.

Step 9: Configure Network Settings

1. Choose VPC

- Default VPC (recommended for beginners)
- Custom VPC (created manually)

2. Choose Subnet

- e.g., `us-east-1a` Availability Zone

3. Auto-assign Public IP

- **Enable** if you want SSH / public access.

4. Security Group (Firewall Settings)

Create a new Security Group:

Name: EC2-Web-SG

Add rules:

Type	Protocol	Port	Source	Purpose
SSH	TCP	22	My IP	Connect to instance
HTTP	TCP	80	0.0.0.0/0	Host website
HTTPS	TCP	443	0.0.0.0/0	Secure website

Step 10: Configure Storage (EBS Volume)

Example:

Volume	Size	Type	Purpose
Root	8–20 GB	gp3	Default OS disk

You can also add:

- Additional volumes
- IOPS configuration
- Throughput settings

Step 11: Review and Launch

Review:

- Region
- AMI
- VPC/Subnet
- Security Group
- Key Pair
- EBS Volume

Click **Launch Instance**.

4. Post-Launch Steps

Step 12: View Your Instance

You will see:

- Instance ID
- Instance State: `running`
- Public IPv4: `52.x.x.x`
- Private IPv4: `172.x.x.x`

Wait until **Status Checks = 2/2 passed**.

5. Connect to the EC2 Instance

Step 13: Connect using SSH (Linux/Mac)

Open terminal:

```
chmod 400 my-key.pem
ssh -i "my-key.pem" ec2-user@PUBLIC_IP
```

Ubuntu AMI username: `ubuntu`

Amazon Linux AMI username: `ec2-user`

Step 14: Connect using SSH (Windows)

Use **Windows Terminal** or **Git Bash**:

```
ssh -i my-key.pem ec2-user@PUBLIC_IP
```

Or use **PuTTY** (convert `.pem` → `.ppk`).

6. Install a Sample Web Server (Optional)

Inside the instance:

```
sudo yum install httpd -y # For Amazon Linux
sudo systemctl start httpd
sudo systemctl enable httpd
echo "Welcome to my EC2 Web Server!" | sudo tee /var/www/html/index.html
```

Visit:

`http://YOUR_PUBLIC_IP/`

7. Terminating the Instance

To avoid charges:

1. Select instance
2. Click **Instance State** → **Terminate**
3. Delete associated EBS volumes if not needed

8. Best Practices

Security

- Never allow SSH (22) from 0.0.0.0/0 otherwise you will end up providing SSH access to everyone
- Use IAM roles instead of storing credentials
- Enable CloudWatch alarms for monitoring

- Stop or terminate unused instances